

Final Project Report Template

1. Introduction

1.1. Project overviews

This project analyzes workforce reduction trends in the technology sector using data on layoffs, market conditions, employee sentiment, and AI adoption. An interactive dashboard helps identify key patterns and provides insights for workforce planning and decision-making.

1.2. Objectives

To analyze workforce reduction trends in the technology sector by examining layoffs across companies, industries, countries, and time periods, and to understand the impact of AI adoption on job displacement and employment patterns.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
Sarah needs to analyse industry-wide layoff trends and compare her company's workforce reduction rates with competitors to make informed workforce planning decisions	An HR Director at a major technology company.	Compare layoff percentages across countries and industries, benchmark my company against competitors, and identify critical hiring	Layoff trends vary significantly across industries and countries, making it	I need reliable, clear, and comparative data to understand where layoffs are highest and where talent shortages	Concerned about workforce stability and the need to make the right talent retention and hiring decisions.

		gaps.	difficult to quickly identify relevant benchmarks and workforce risks	may affect business continuity	
Marcus needs to understand how AI adoption is related to job displacement and identify which technology-sector roles may face higher replacement risk.	An AI Ethics Researcher studying the impact of AI adoption on employment	Analyse AI adoption levels, job displacement, layoff trends, and the vulnerability of different job roles across companies.	Analyse AI adoption levels, job displacement, layoff trends, and the vulnerability of different job roles across companies.	AI adoption is changing workforce requirements differently across companies and job roles, creating complex patterns in employment and job security.	Concerned about understanding the ethical and employment implications of AI-driven workforce changes.

2.2. Project Proposal (Proposed Solution)

Project Overview	
Objective	To analyze workforce reduction trends in the technology sector by examining layoffs across companies, industries, countries, and time periods, and to understand the impact of AI adoption on job displacement and employment patterns.
Scope	To analyze technology-sector layoffs across companies, industries, countries, and time periods, identify workforce reduction patterns, and study the impact of AI adoption on job displacement and employment trends.
Problem Statement	
Description	This project analyzes workforce reduction trends in the technology sector by studying layoffs across companies, industries, countries, and time periods. It also examines how AI adoption affects job displacement and employment patterns.
Impact	The project provides data-driven insights into technology-sector layoffs and AI-driven job displacement, helping organizations understand workforce trends, improve workforce planning, and identify areas requiring talent retention or hiring.
Proposed Solution	
Approach	The project uses a data-driven dashboard approach to analyze technology-sector workforce reductions. Data is collected and cleaned, then analyzed based on company, industry, country, time period, layoffs, and AI adoption levels . Interactive visualizations

	are used to identify layoff trends, compare sectors, and understand the relationship between AI adoption and job displacement .
Key Features	The project analyzes layoff trends across companies, industries, and countries, while examining the relationship between AI adoption and job displacement. It provides an interactive dashboard with filters and visualizations to identify workforce reduction patterns and job-risk trends.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	NVIDIA V100 GPUs
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy
Development Environment	IDE, version control	Jupyter Notebook, Git
Data		
Data	Source, size, format	Kaggle dataset, 10,000 images

2.3. Initial Project Planning

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Team members	Priority	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Data collection	USN-1	Downloading Dataset	2	Sahil Pooniya	High	12 August 2026	18 August 2026
Sprint-1	Data collection	USN-2	Loading the dataset	1	Sahil Pooniya	Low	12 August 2026	18 August 2026
Sprint-1	Data preparation	USN-3	Clean the data	3	Sahil Pooniya	High	12August 2026	18 August 2026
Sprint-1	Data visualization	USN-4	Developing visualization	5	Sahil Pooniya	Medium	12 August 2026	18 August 2026
Sprint-2	Dashboard	USN-5	Creating Dashboard	3	Sahil Pooniya	High	19 August 2026	25 August 2026
Sprint-2	Story	USN-6	Creating story	3	Sahil Pooniya	Medium	19 August 2026	25 August 2026
Sprint-2	Documentation and Recording	USN-7	Creating Demo	3	Sahil Pooniya	High	19August 2026	25 August 2026
Sprint 2	Documentation and Recording	USN-8	Creating Documents	5	Sahil Pooniya	High	19 August 2026	25 August 2026

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

Data Collection Plan

Section	Description
Project Overview	This project provides a comprehensive analysis of workforce reduction in the technology sector by studying layoff trends across companies, industries, countries, and time periods. It also examines the impact of AI adoption on job displacement and provides data-driven insights through an interactive dashboard.
Data Collection Plan	Data will be collected from reliable technology and employment sources covering company layoffs, industry, country, date, workforce reduction percentage, AI adoption, and job roles. The collected data will be cleaned, organized, and analyzed to identify layoff trends, workforce reduction patterns, and the impact of AI adoption on employment.
Raw Data Sources Identified	Data will be collected from reliable technology and employment sources covering company layoffs, industry, country, date, workforce reduction percentage, AI adoption, and job roles. The collected data will be cleaned, organized, and analyzed to identify layoff trends, workforce reduction patterns, and the impact of AI adoption on employment.

Raw Data Sources

Source Name	Description	Location/URL	Format	Size	Access Permissions
Dataset 1	UNESCO	https://databrowser.uis.unesco.org/resources/bulk?utm_source=chatgpt.com	CSV	1GB	Public
Dataset 2	OECD	https://www.oecd.org/en/data/datasets/oecd-DE.html?utm_source=chatgpt.com	Excel	1GB	Public
Dataset 3	Smart bridge site	https://thesmartbridge.com/	Excel	1 GB	Public

3.2. Data Quality Report

Data Source	Data Quality Issue	Severity	Resolution Plan
Dataset 1	Some missing numbers.	Moderate	Filter the data and add missing numbers
Dataset 2	Reports related to Comprehensive Analysis of Technology Sector Workforce Reduction are separated.	High	Make a single excel file containing all reports.
Dataset 3	Some values of dataset are different from other dataset	medium	Compare with other dataset and find authentic data from official websites.

3.3. Data Exploration and Preprocessing

Section	Description
Data Overview	Data set is related to Comprehensive Analysis of Technology Sector Workforce Reduction

Data Cleaning	Execute data cleaning workflows to impute missing entries, deduplicate records, and rectify data anomalies
Data Transformation	Data manipulation: filtering, sorting, pivoting, and creating calculated columns.
Data Type Conversion	Rectifying Datatype
Column Splitting and Merging	Split or merge columns as needed.
Data Modeling	Define relationships between tables and construct measures.
Save Processed Data	Save the cleaned and processed data for future use.

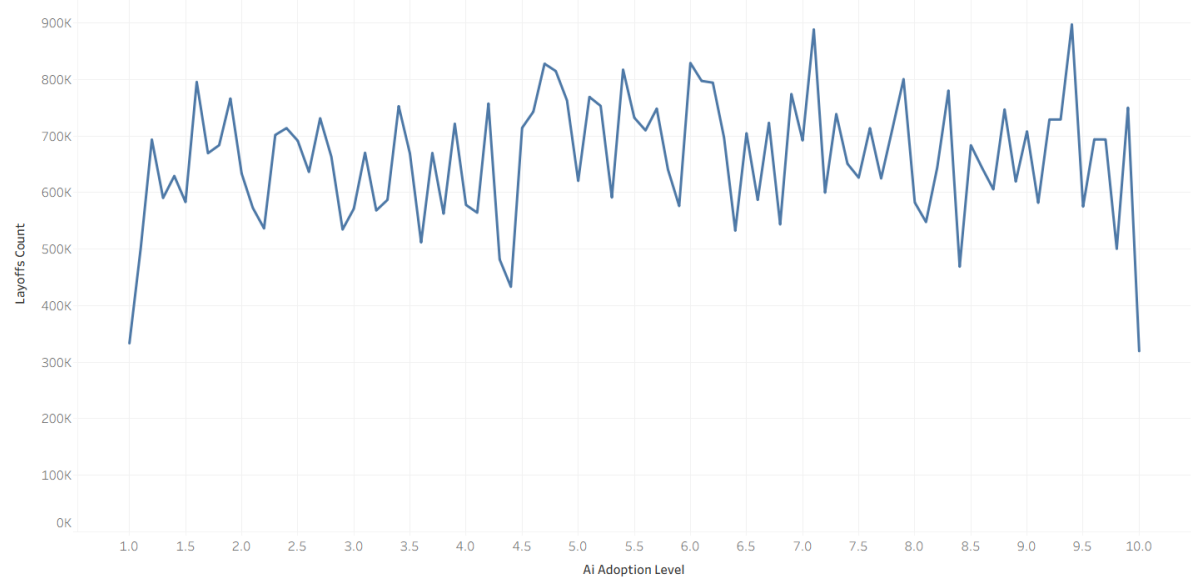
4. Data Visualization

4.1. Framing Business Questions

Q.1 What is the relationship between AI adoption levels and workforce reductions across technology companies?

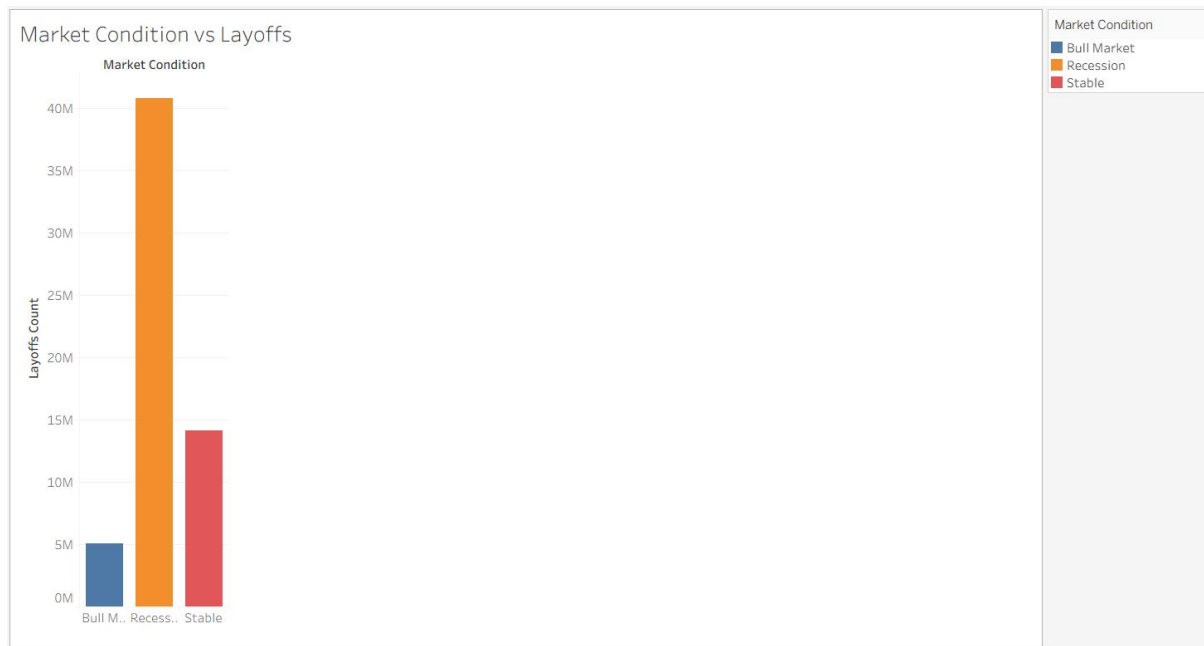
Ans. Higher AI adoption is associated with increased workforce reduction in some technology companies.

AI Adoption vs Layoffs Impact



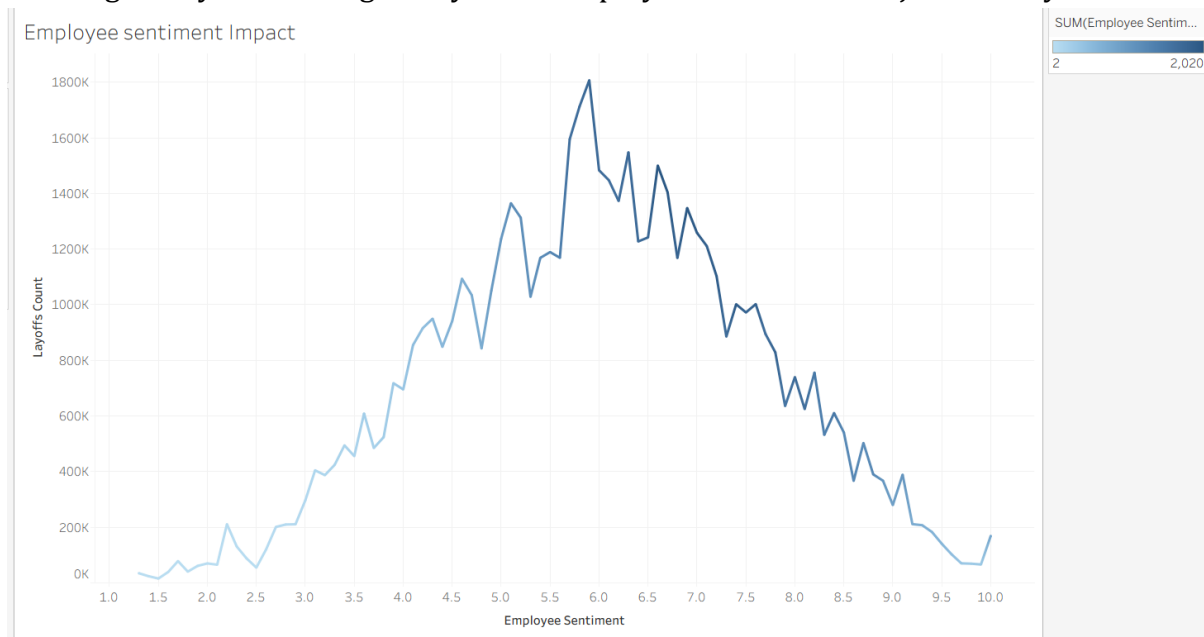
Q.2 How do market conditions influence layoffs?

Ans Unfavorable market conditions generally lead to higher workforce reductions



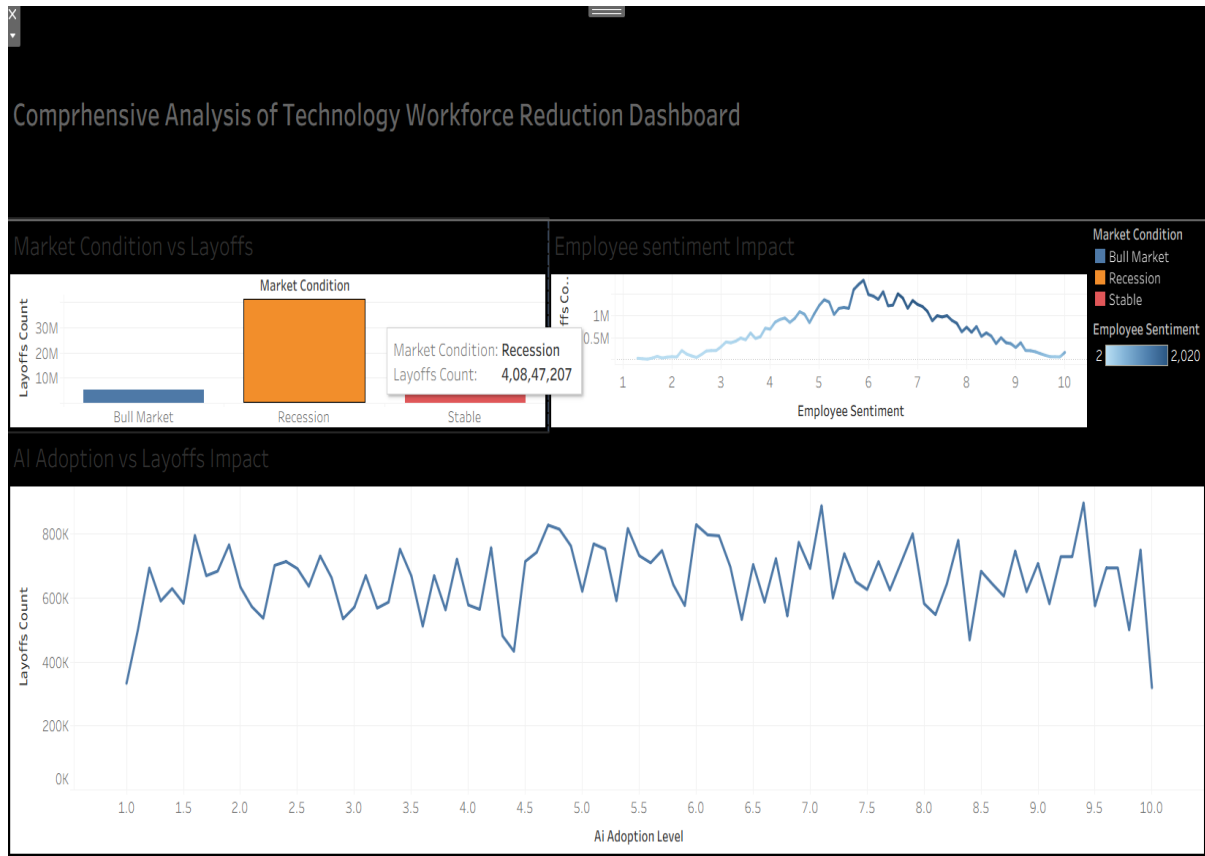
Q.3 What impact do layoffs have on employee sentiment?

Ans. Higher layoffs can negatively affect employee sentiment and job security



Q.4 What are the major factors driving workforce reduction?

Ans. AI adoption, market conditions, and changing workforce requirements are key factors.

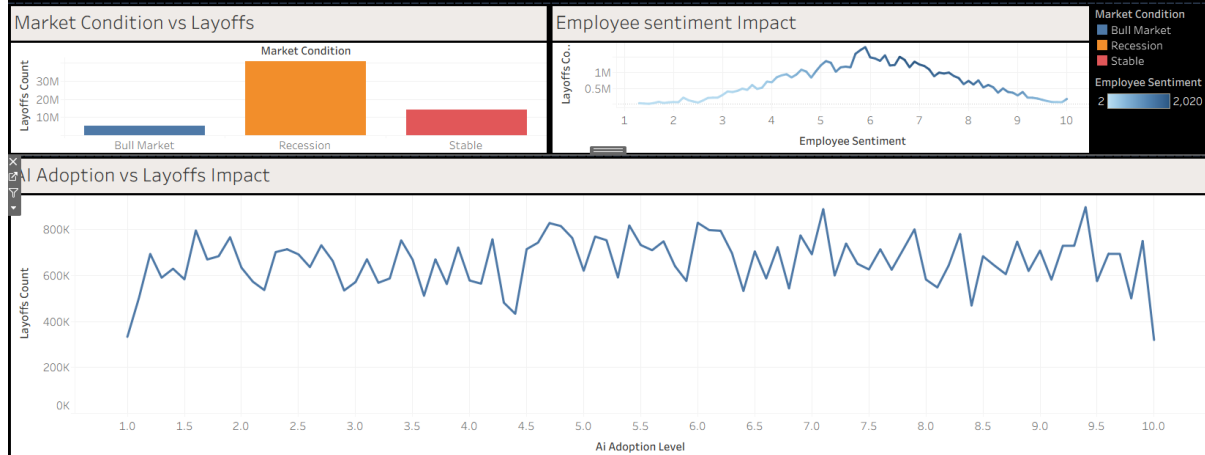


5. Dashboard

5.1. Dashboard Design File

Activity 1: Interactive and visually appealing dashboard

Comprehensive Analysis of Technology Workforce Reduction Dashboard



Outcomes

- **Market Conditions:** Layoffs are highest during **recession**, compared with stable and bull-market conditions.
- **Employee Sentiment:** Layoff levels show noticeable variation across different employee sentiment levels.
- **AI Adoption:** Layoffs fluctuate across AI adoption levels, with **no simple linear relationship** between AI adoption and layoffs.
- **Overall Outcome:** The dashboard highlights how **market conditions, employee sentiment, and AI adoption** are associated with workforce reduction trends in the technology sector.

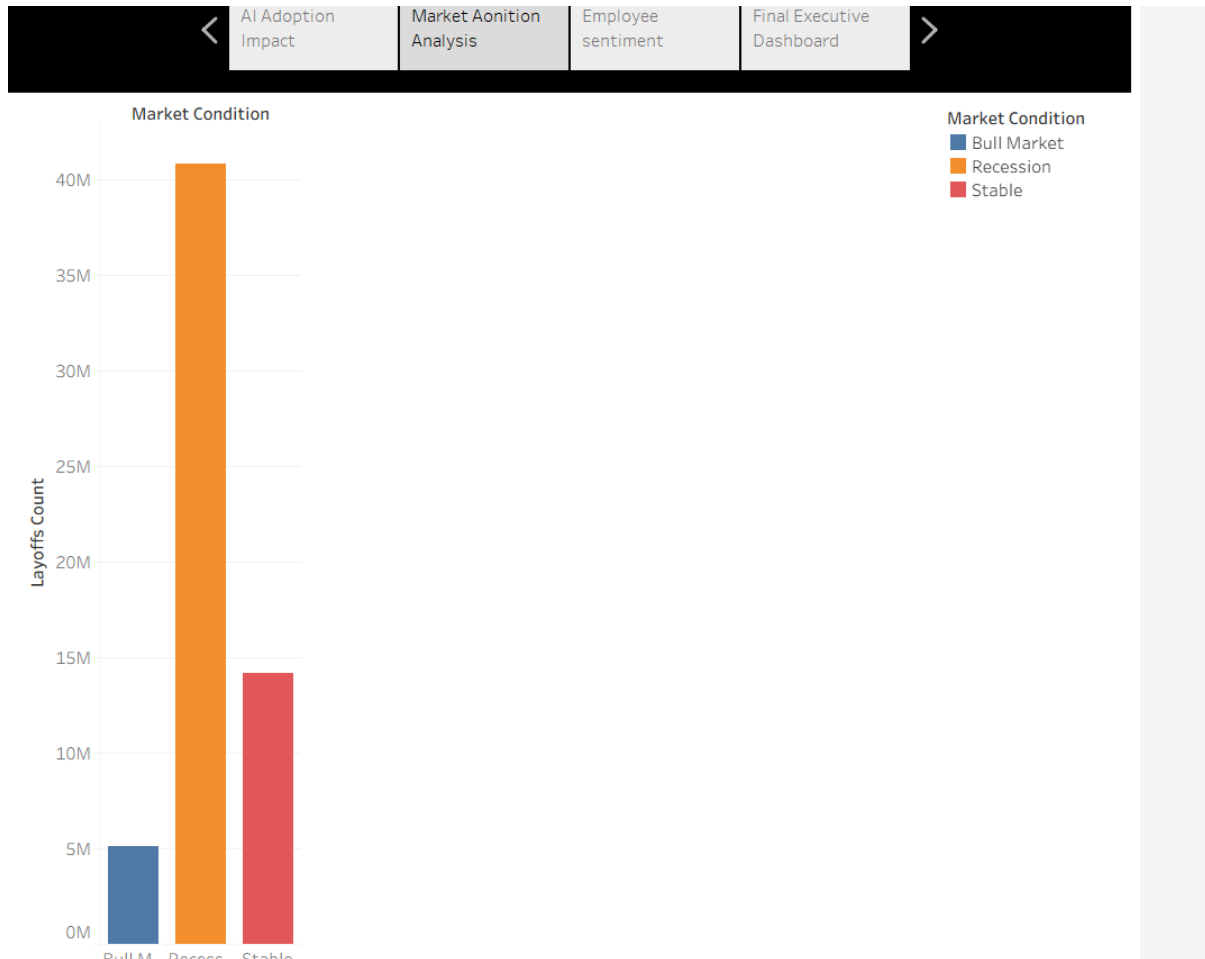
Activity 2

Publish Dashboard on Tableau Public and Paste the Dashboard Public link below:

https://public.tableau.com/views/TechnologyWorkforcereductionanalysis2/Dashboard1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

6. Report

6.1. Story Design File



Observations

- **Recession periods show the highest layoffs** compared with stable and bull-market conditions.
- **Employee sentiment varies with layoff levels**, showing changes in workforce morale and job security.
- **AI adoption and layoffs fluctuate together**, but the dashboard does not show a clear linear relationship.
- **Workforce reduction is influenced by multiple factors**, including market conditions, AI adoption, and employee sentiment.

- The technology sector shows **significant variation in layoffs across different AI adoption levels.**

Activity 2:

Publish Story on Tableau Public and Paste Public link below

https://public.tableau.com/views/TechnologyWorkforcereductionanalysis2/Story1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

7. Performance Testing

7.1 Utilization of Data filters

Filters are used to get required visualizations.

8. Conclusion/Observation

The project concludes that technology-sector workforce reduction is influenced by multiple factors, particularly market conditions, employee sentiment, and AI adoption. Recession periods show higher layoffs, while AI adoption shows varying workforce impacts. The dashboard provides useful insights for workforce planning, talent management, and understanding changing employment trends.

9. Future Scope

The project can be expanded by adding real-time layoff data, more countries and industries, job-role level analysis, and AI-based prediction models to forecast future workforce reduction trends and support better workforce planning.

10. Appendix

10.2. GitHub

<https://github.com/sahilpooniya/Technology-Workspace-Analysis>

& Project Demo Link

<https://drive.google.com/file/d/1vWR1kWmE4G0ruoc3QhCAHE4FpVSacP2i/view?usp=sharing>